Lecture 13: EMF and Resistors

Pre-Lecture Worksheet

Complete the following worksheet in preparation for the next lecture. The following questions correspond to material covered in chapter 25. Feel free to read chapter 25 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

1) EMF stands for:

Electromotive force

2) List 3 common sources of EMF.

Cell Phones, wifi routers, computers

3) Define EMF and draw the symbol.

ε energy supplied per unit electric charge by a source

4) (True or false) EMF accounts for the internal resistance of the battery?

F

5) What is ‘r’ in the following equation? What is V?

V = voltage

r = resistance

$$V_{ba} = \varepsilon - Ir$$

6) Resistance is one way of extracting work from a system. Answer the following questions about each orientation.

(a) label the charges on the battery

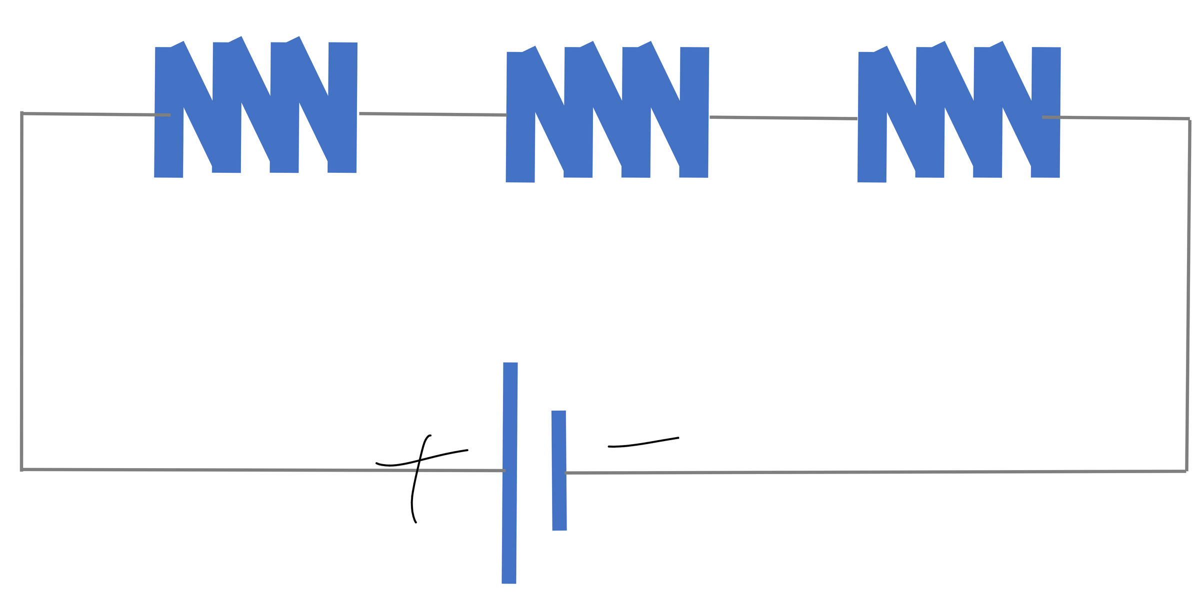
(b) label whether the system is parallel or series

(c) write the appropriate equation for the equivalent resistance of the system

(d) state whether the system adds resistance and/or voltage.

(e) how does resistance addition differ from capacitance?

A

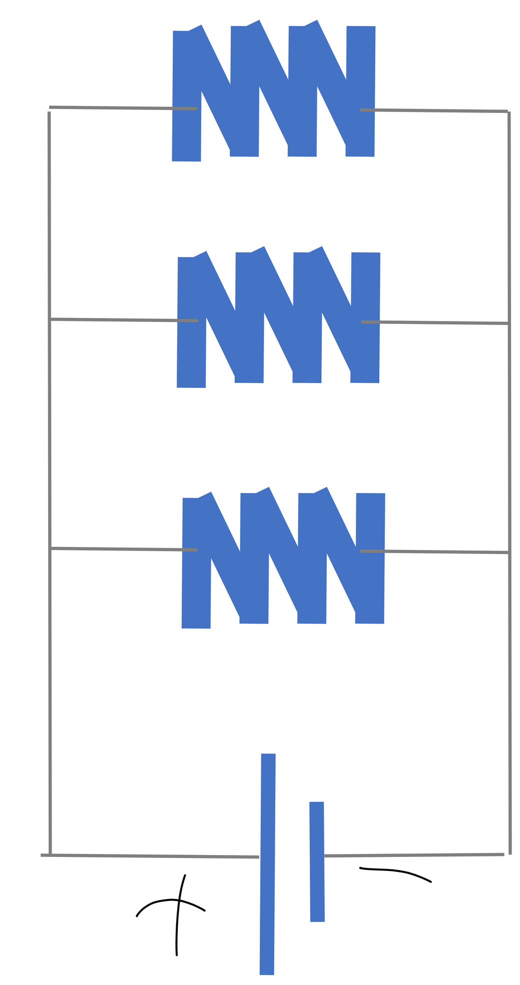


series

$R_{eq} = R_1 + R_2 + R_3$

adds Resistance

B



parallel

$R_{eq} = R_1^{-1} + R_2^{-1} + R_3^{-1}$

adds voltage

equations Differ, capacitors in series use the inverse, while Resistance simply sums.